

4. (a) Explain the conditions that are required of the light sources, for a two-source interference pattern to be observed. Include examples of when the requirements would and would not be met. [6 QER]

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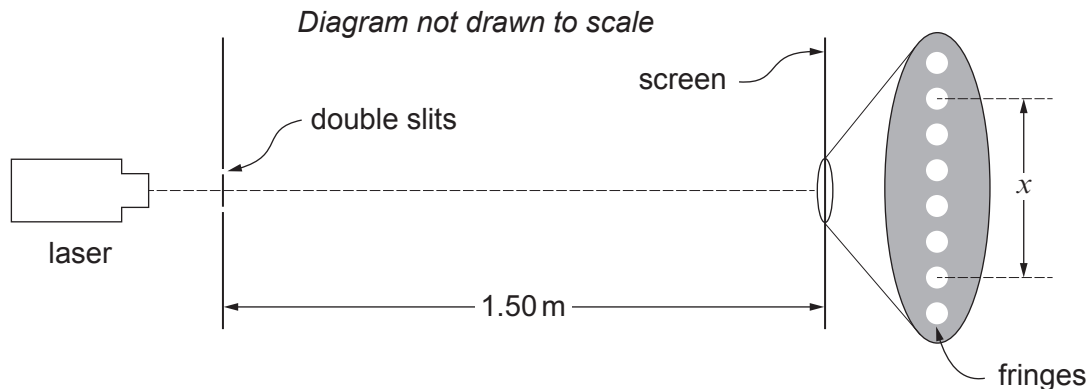
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(b) A Young's fringes experiment is set up as shown.



(i) Six students measure the distance x , obtaining these results:

6.5 mm 6.3 mm 6.9 mm 6.9 mm 6.7 mm 6.4 mm

Calculate the mean value for fringe separation (the separation of the centres of **neighbouring** bright fringes), together with its **percentage** uncertainty. [4]

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(ii) The uncertainty in the distance of 1.50 m shown in the diagram is negligible. The separation between the centres of the slits is given by the makers of the double slit as $0.60 \text{ mm} \pm 5\%$. Calculate a value for the wavelength of the light from the laser, along with its **percentage** uncertainty. [3]

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QUESTION CONTINUES ON NEXT PAGE



- (c) The laser beam is now shone normally on to a diffraction grating with centres of slits separated by 1500 nm . Second order beams emerge at angles of 45° to the normal.

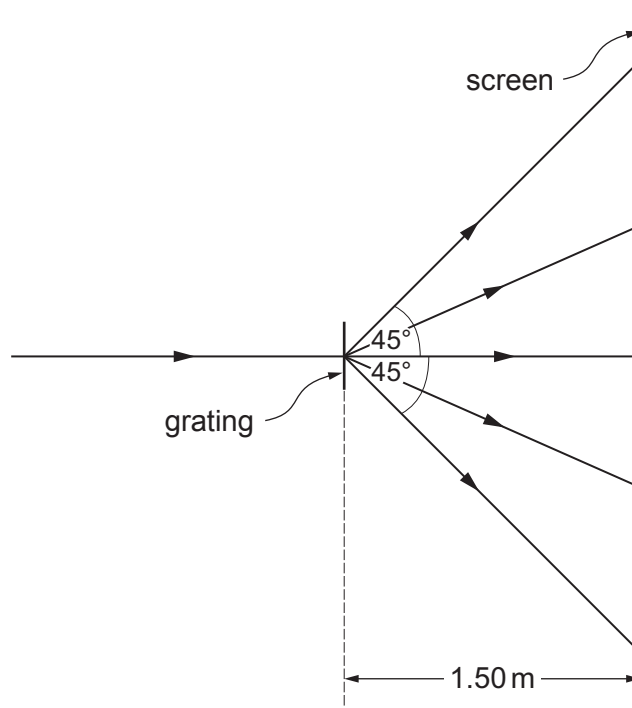


Diagram not drawn to scale

- (i) Determine a value for the wavelength of the laser light from the data. [2]

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- (ii) The beams from the grating strike the screen as shown opposite. The pattern on the screen may be compared with the pattern of fringes in part (b) (Young's experiment). There are fewer bright spots in the case of the grating. State two other differences between the patterns **and** state briefly why each difference occurs. [4]

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